HESHAM ABOUKEILA

ACADEMIC CURRICULUM VITAE

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RESEARCH PROFILE

Chemical engineer investigating structure, processing, and performance relationships in sustainable polymers. Research spans biobased and biodegradable polyester films, crystallization, polymer blends, chemical recycling, gas separation membranes, and surfactant formulations. Inventor on U.S. Patent No. 12,630,710 B1 and author or coauthor of 13 peer-reviewed journal articles.

EDUCATION

Ph.D., Chemical Engineering | University of Oklahoma **Dec 2024**

Dissertation: Synthesis and characterization of biobased polyesters to replace LDPE/LLDPE in flexible film packaging.

Advisors: Prof. Brian P. Grady and Prof. John Klier.

M.Sc., Chemical Engineering | University of Bologna **Mar 2018**

Thesis: Production and characterization of hybrid polymeric membrane for carbon dioxide capture.

Advisors: Prof. Marco Giacinti Baschetti and Prof. Liyuan Deng.

B.Sc., Gas and Petrochemicals Engineering | Alexandria University **Jul 2014**

Capstone project: Effect of using different types of packing on removal of carbon dioxide by Benfield solution.

Advisor: Prof. Nevin Kamal Amin Abbas.

RESEARCH APPOINTMENTS AND CONTRIBUTIONS

Postdoctoral Research Associate | University of Oklahoma **Jan 2025 - Jan 2026**

- Led a Church & Dwight funded study comparing three nonionic surfactants in potassium carbonate and sodium chloride systems; examined an accessible HLD span of approximately 8 units with potassium carbonate versus approximately 4.5 units available with sodium chloride, and identified detergency optima reaching 90% soil removal.
- Characterized surfactants derived from post-consumer recycled HDPE using critical micelle concentration, surface and interfacial tension, foam stability, and calcium tolerance measurements; demonstrated CMC values as low as 0.08 wt% and calcium tolerance of 221 ppm in the investigated systems.

Graduate Research and Teaching Assistant | University of Oklahoma **Aug 2021 - Dec 2024**

- Engineered and evaluated more than 60 biobased biodegradable polyester formulations for flexible packaging, integrating synthesis, crystallization, processing, and structure-property analysis.
- Investigated ionomer nucleation of PPeAT, reducing crystallization half time from approximately 28 minutes to less than one minute under the investigated conditions.
- Produced more than 20 m of polyester and polyethylene films by laboratory-scale blown-film extrusion and evaluated thermal, rheological, mechanical, and barrier properties. PPeAT blown films demonstrated approximately three times the modulus of the commercial PBAT benchmark and lower oxygen and carbon dioxide permeability than PBAT and LLDPE under the reported test conditions.
- Contributed to polyester blending and closed-loop chemical recycling studies, including amine-catalyzed methanolysis and repolymerization of PPeAT with properties comparable to the original material.

Graduate Research Assistant | Norwegian University of Science and Technology **Sep 2017 - Mar 2018**

- Investigated Nafion/PEGDME hybrid and TESET multiblock gas-separation membranes under the supervision of Prof. Liyuan Deng, relating additives, conditioning and humidity to morphology and CO2 transport. Nafion/PEGDME systems achieved up to 36 times higher dry-state CO2 permeability and CO2/N2 selectivity above 30 under humid conditions.

Graduate Research Assistant | University of Bologna **Mar 2016 - Aug 2017**

- Investigated Aquivion ionomer membranes under the supervision of Prof. Marco Giacinti Baschetti, examining humidity and operating-condition effects on CO2 and hydrocarbon permeation for gas separation.

RESEARCH METHODS AND TECHNICAL EXPERTISE

Processing: extrusion, compounding, film blowing, compression molding. Characterization: FTIR, DSC, TGA, SAXS, XRD, SEM, DMA. Performance: tensile testing, shear and extensional rheology, OTR/WVTR, gas permeation. Surface and colloid science: CMC, surface/interfacial tension, contact angle, bubble pressure, foam analysis. Software: Aspen Plus, Aspen HYSYS, MATLAB, Symmetry, Origin, Python.

PEER REVIEWED JOURNAL PUBLICATIONS

1. **Hesham Aboukeila**, Marlee Leonard, Willow Wise, Steven T. Adamy, Robert T. Zehr, Dave Sabatini, Jeffrey H. Harwell, Brian P. Grady. (2026). Impact of Potassium Carbonate on the Detergency Behavior of Nonionic Surfactants. Journal of Surfactants and Detergents, 29(5), 849-857. DOI

2. **Hesham Aboukeila**, Mahdi Mazaheri, Onkar Singh, Emily Edwards, George W. Huber, John Klier, Brian P. Grady. (2026). Structure–Property Relationships in PLA–PBAT and PLA–PPeAT Blends: Toward Stronger Biobased Biodegradable Packaging Materials. Journal of Materials Science, 61(24), 17594-17608. DOI

3. **Hesham Aboukeila**, Onkar Singh, Mahdi Mazaheri, Julia B. Curley, Henry L. Cater, Gregg T. Beckham, John Klier, Brian P. Grady. (2026). Closed-Loop Recycling of Poly(pentylene adipate-co-terephthalate) via Amine-Catalyzed Methanolysis. ChemSusChem, 19(9), e70664. DOI

4. **Hesham Aboukeila**, Emily Edwards, Onkar Singh, John Klier, Brian P. Grady. (2026). Enhanced Crystallization of Poly(pentylene adipate-co-terephthalate) via Ionomer Nucleation: Toward Biodegradable Flexible Packaging. Polymer Engineering & Science, 66(3), 1786-1795. DOI

5. Houqian Li, Brandon W. Tipton, **Hesham Aboukeila**, Enner A. Mendoza, Abdulrahman Alzailaie, Tianwei Yan, Clark R. Landis, Javen S. Weston, Brian P. Grady, George W. Huber. (2025). Upcycling of Waste Plastics into Carboxylic Acids for Biodegradable Surfactants. Angewandte Chemie International Edition, 64(50), e202517471. DOI

6. **Hesham Aboukeila**, Onkar Singh, Mahdi Mazaheri, Emily Edwards, Mason A. Rhue, John Klier, George W. Huber, Brian P. Grady. (2026). Film Blowing of Biobased Biodegradable Polyesters: Poly(pentylene adipate-co-terephthalate) and Poly(dodecylene furanoate). Macromolecular Materials and Engineering, 311(1), e00233. DOI

7. **Hesham Aboukeila**, Eswara Rao Chokkapu, Hang-Fei Tu, Onkar Singh, Wilfred T. Diment, Shu Xu, Meltem Urgun-Demirtas, John Klier, George W. Huber, Brian P. Grady, Eugene Y.-X. Chen. (2025). Biobased Poly(dodecylene Furanoate) with Inherent Advantages in Performance and Circularity. ChemSusChem, 18(17), e202501080. DOI

8. **Hesham Aboukeila**, Onkar Singh, John Klier, George W. Huber, Brian P. Grady. (2024). Synthesis and Characterization of Biobased Copolyesters Based on Pentanediol: (2) Poly(pentylene adipate-co-terephthalate). Polymer Engineering & Science, 64(10), 4746-4759. DOI

9. Onkar Singh, **Hesham Aboukeila**, John Klier, George W. Huber, Brian P. Grady. (2024). Synthesis and Characterization of Biobased Copolyesters Based on Pentanediol: (1) Poly(pentylene dodecanoate-co-furandicarboxylate). Polymer Engineering & Science, 64(10), 4935-4946. DOI

10. Sergio A. Pineda-Castillo, Tanner L. Cabaniss, **Hesham Aboukeila**, Brian P. Grady, Hyowon Lee, Bradley N. Bohnstedt, Yingtao Liu, Chung-Hao Lee. (2023). Toward the Development of a Shape Memory Polymer for Individualized Endovascular Therapy of Intracranial Aneurysms Using a 3D Printing/Leaching Method. Advanced Engineering Materials, 25(23), 2300683. DOI

11. Zhongde Dai, Jing Deng, **Hesham Aboukeila**, Jiaqi Yan, Luca Ansaloni, Kenneth P. Mineart, Marco Giacinti Baschetti, Richard J. Spontak, Liyuan Deng. (2019). Highly CO_2-Permeable Membranes Derived from a Midblock-Sulfonated Multiblock Polymer after Submersion in Water. NPG Asia Materials, 11(1), 53. DOI

12. Zhongde Dai, **Hesham Aboukeila**, Luca Ansaloni, Jing Deng, Marco Giacinti Baschetti, Liyuan Deng. (2019). Nafion/PEG Hybrid Membrane for CO_2 Separation: Effect of PEG on Membrane Micro-structure and Performance. Separation and Purification Technology, 214, 67-77. DOI

13. Luca Olivieri, **Hesham Aboukeila**, Marco Giacinti Baschetti, Diego Pizzi, Luca Merlo, Giulio C. Sarti. (2017). Humid Permeation of CO_2 and Hydrocarbons in Aquivion Perfluorosulfonic Acid Ionomer Membranes, Experimental and Modeling. Journal of Membrane Science, 542, 367-377. DOI

PATENT

Aboukeila, H.; Grady, B. P.; Klier, J. 1,5-Pentanediol-Based Polyester Compositions with Enhanced Crystallization Rates and Methods of Manufacture. U.S. Patent No. 12,630,710 B1 (2026).

Related U.S. provisional application: No. 63/554,363, filed February 16, 2024. Patent

TEACHING EXPERIENCE

Chemical Engineering Instructor | American University of the Middle East **Sep 2019 - Aug 2021**

Egaila, Kuwait. Eight distinct courses across four semesters. Section enrollment was approximately 30 students before COVID-19 and 60 during online instruction; teaching loads varied by semester. Spring 2020 transitioned to online instruction midway through the semester.

Semester	Course	Sections	Role and responsibilities
Fall 2019	CHE 205: Chemical Engineering Calculations	2	Teaching assistant; tutorials
Fall 2019	CHE 306: Design of Staged Separation Processes	2	Teaching assistant; tutorials
Fall 2019	CHE 377: Momentum Transfer	3	Laboratory instructor
Spring 2020	CHE 348: Chemical Reaction Engineering	4	Tutorials and laboratory instruction
Spring 2020	CHE 377: Momentum Transfer	4	Tutorials and laboratory instruction
Spring 2020	CHE 300: Chemical Engineering Junior Seminar - Engineering Ethics	1	Instructor
Fall 2020	CHE 306: Design of Staged Separation Processes	3	Teaching assistant; tutorials
Fall 2020	CHE 377: Momentum Transfer	3	Tutorials and laboratory instruction
Fall 2020	CHE 300: Chemical Engineering Junior Seminar - Engineering Ethics	3	Sole instructor and course moderator
Fall 2020	CHE 435: Chemical Engineering Laboratory	3	Laboratory instructor
Spring 2021	CHE 300: Chemical Engineering Junior Seminar - Engineering Ethics	3	Sole instructor and course moderator
Spring 2021	CHE 378: Heat and Mass Transfer	4	Tutorials and laboratory instruction
Spring 2021	CHE 211: Introductory Chemical Engineering Thermodynamics	2	Teaching assistant; tutorials

Graduate Teaching Assistant | University of Oklahoma **Fall 2021 - Spring 2022**

Semester	Course	Responsibilities
Fall 2021	CH E 3473: Chemical Engineering Thermodynamics	Tutorials and worked examples; junior level
Spring 2022	CH E 3333: Separation Processes	Aspen Plus tutorials; junior level

CURRICULUM DEVELOPMENT AND RESEARCH MENTORING

- Rebuilt instructional materials for CHE 300 and independently delivered the course as sole instructor and moderator in Fall 2020 and Spring 2021.
- Proposed and recorded step-by-step experimental demonstrations during COVID-19 remote instruction. Supplied measured results and assigned laboratory reports covering theory, experimental procedures, calculations and conclusions. Coordinated more than ten laboratory tour and equipment introduction videos.
- Mentored two undergraduate and two doctoral researchers; supported experimental training and onboarding of more than 20 researchers in laboratory methods and polymer research.

CONFERENCE PRESENTATIONS

- Oral presentation: Advantages and Disadvantages of Using Salts Other Than NaCl in Detergency Studies. Institute for Applied Surfactant Research, Norman, OK, November 11-12, 2025.
- Poster: Upcycling of Waste Plastic into Carboxylic Acids for Biodegradable Surfactants. Institute for Applied Surfactant Research, Norman, OK, November 11-12, 2025.
- Oral presentation: Synthesis and Characterization of Biobased Copolyesters Based on Pentanediol: Poly(pentylene adipate-co-terephthalate). AIChE Annual Meeting, San Diego, CA, October 27-31, 2024.
- Invited talk: Synthesis and Characterization of Biobased Copolyesters Based on Furandicarboxylic Acid: Poly(dodecamethylene furandicarboxylate). ACS Fall Meeting, Denver, CO, August 18-22, 2024.
- Poster: Synthesis and Characterization of Biobased Copolyesters Based on Pentanediol: Poly(pentylene adipate-co-terephthalate). ACS Fall Meeting, Denver, CO, August 18-22, 2024.
- Poster: Synthesis and Characterization of Biobased Copolyesters Based on Furandicarboxylic Acid: Poly(dodecamethylene furandicarboxylate). SPE ANTEC, St. Louis, MO, March 4-7, 2024.
- Poster: Synthesis and Characterization of High Molecular Weight Branched Poly(pentylene adipate-co-terephthalate). ACS Southwest Regional Meeting, Oklahoma City, OK, November 15-18, 2023.
- Poster: Characterization of High Molecular Weight Poly(pentylene adipate-co-terephthalate). ACS Fall Meeting, San Francisco, CA, August 13-17, 2023.
- Poster: Effect of Modified Silica on the Glass Transition Temperature of Polyisobutylene. SPE ANTEC, Denver, CO, March 27-30, 2023.

HONORS AND AWARDS

- Eastman Chemical Student Award in Applied Polymer Science Finalist, ACS PMSE Division (2024)
- Dissertation Excellence Award, Gallogly College of Engineering (2024)
- Conference/Travel Grant Award, Graduate Student Senate (2024)
- Graduate Student Travel Award, Gallogly College of Engineering (2024)
- Ken Starling Endowed Scholarship, School of Sustainable Chemical, Biological and Materials Engineering (2024)
- Armstrong Memorial Scholarship, Gallogly College of Engineering (2024)
- Armijo-Tucei Chemical Engineering Scholarship, Gallogly College of Engineering (2024)
- ANTEC Travel Award, Society of Plastics Engineers (2024)
- Judith K. Pearson Memorial Engineering Award, Gallogly College of Engineering (2023)

ACADEMIC SERVICE AND PROFESSIONAL AFFILIATIONS

- President, Society of Plastics Engineers, University of Oklahoma Student Chapter (2022 - 2024); organized an annual Chevron Phillips Chemical industry visit for approximately 15 attendees.
- Chair, Academic Affairs Committee, Graduate Student Senate (2022 - 2024); redesigned the awards evaluation rubric, reducing evaluation time by approximately half.
- Reviewer, Polymer Engineering & Science (2025 - Present)
- Reviewer, Journal of Polymer Engineering (2022 - Present)
- Officer, Chemical Engineering Graduate Student Society (2021 - 2024)
- Member, American Chemical Society (2023 - Present)
- Member, American Institute of Chemical Engineers (2019 - Present)
- Erasmus Exchange Student, Norwegian University of Science & Technology (Sep 2017 - Mar 2018).

INDUSTRY RESEARCH COLLABORATION

Pyran, Inc. (2024-2025): Performed polymer synthesis and characterization using Pyran-supplied 1,5-pentanediol; prepared technical findings provided to the company in November 2024 and participated in subsequent technical correspondence.

FEDERALLY SUPPORTED RESEARCH AND TECHNICAL REPORTS

Designing Recyclable Biomass-Based Polyesters. U.S. Department of Energy final technical report, Award DE-EE0009305 (2025). Principal investigator: Prof. George Huber, University of Wisconsin–Madison. Contributed to University of Oklahoma polymer synthesis, characterization, and flexible-film research within the collaborative project.

ADDITIONAL PROFESSIONAL EXPERIENCE

Sales & Service Engineer | System Technique | Cairo, Egypt **Jul 2018 - Jul 2019**

Provided technical troubleshooting, client training, and engineering support for water-treatment installations. Expanded the active client base from four to twelve accounts and met the annual sales target within six months.

LANGUAGES

Arabic (Native) | English (Fluent) | Italian (Beginner)